

Project 9 : Sound Module

Introduction:

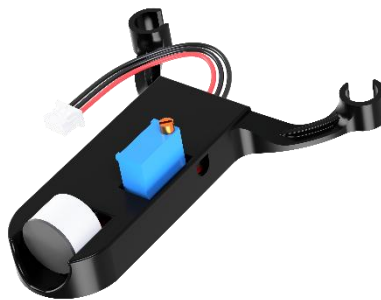
Pluto X, One of the add-ons is Sound Sensor. This is a type of sensor which detects a sound from the surrounding.

Challenge:

Build a program for Pluto drone, which on application of Loud sound on sensor, the drone takes off automatically

Add-On used:

A sound sensor is defined as a module that detect sound waves through its intensity and converting it to electrical signals. The sound sensor consists of an in-built capacitive microphone, a peak detector, and an amplifier that's highly sensitive to sound.



Connection:

Connect the Buzzer_module on the ADC port on Primus V5/X2 board

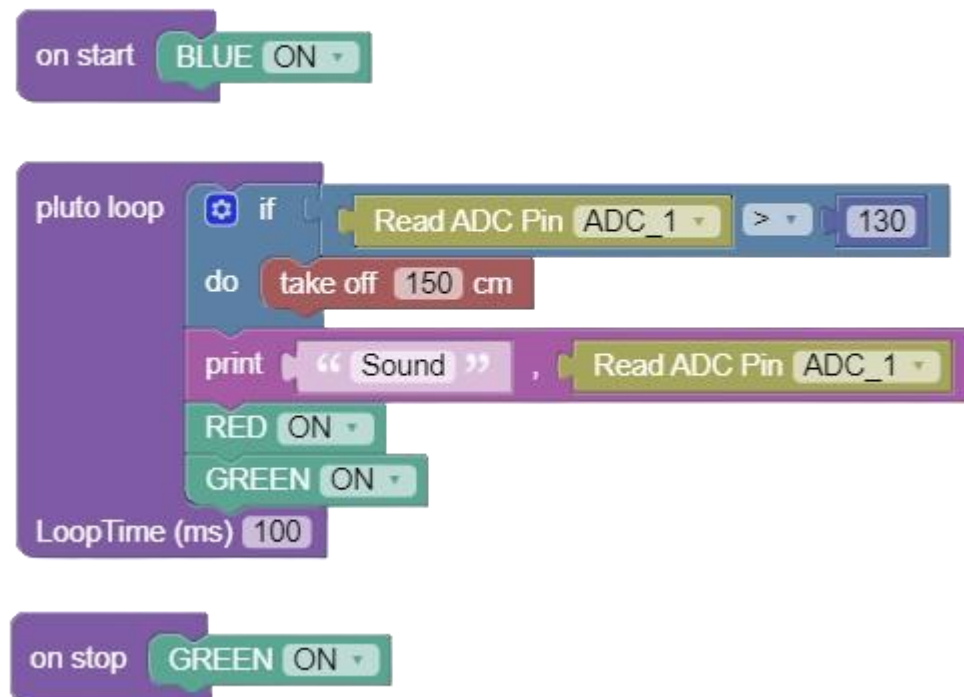
Primus V5/X2 pin	Peripherals
VCC on ADC port	VCC on Sound Sensor
GND on ADC port	GND on Sound Sensor
Signal on ADC port	A0 on Sound Sensor

Approaching the Problem:

We constantly monitor the ADC port pin input data. When the input goes above a specified threshold, we take decision to take off or not.

Solution:

Select Receiver Mode **Rx_ESP**



Explanation of Solution:

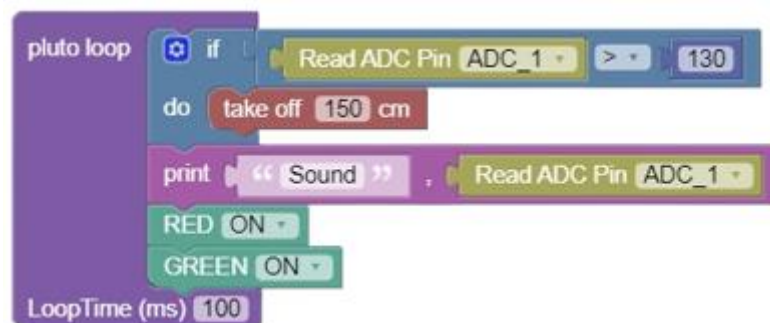
On Start Loop:

We turn On blue Led.



Pluto Loop:

Sound sensor returns the value in terms of analog values from 0 to 4096. but under normal circumstances, it returns values from 115 to 125, when there is no substantially loud sound/vibration. When it occurs the input levels crosses 130 magnitude level. The sensitivity of sensor can be changed by adjusting the potentiometer on the sensor. The values obtained here are experimentally noted, you are free to tweak them. Upon crossing the threshold, the drone takes off, until then it stays low.



On stop Loop: Green led turns ON.





Fig: Sound sensor add-on Pluto

Try further!

- Utilise a different type of sound sensor.
- Make audio reactive Pluto.
- Try adding a microphone to pluto, that remotely records sounds of environment.